

Task 3

Compose Lab

Overview

In this section, MongoDB is used for the persistent storage of data communicated between a RESTful service and a gRPC service. All services including the container hosting MongoDB are defined and run together using docker-compose.yml file. The existing framework and functions from Tasks 1 and 2 are modified for this task. The service/container names, their dependencies, their ports, and their respective networks are described in the .yml file. The CRUD functions can be sent as commands via the RESTful service to the gRPC server, which accesses MongoDB for persistent storage. The RESTful service also includes a circuit breaker which breaks the connection in case of failures.

Github Repository: <https://github.com/bavanmendon/dsa-practicals/tree/master>

Preparation of .yml compose file

This file is the root of the compose implementation. Three containers are defined: one for MongoDB implementation, one for gRPC server, and one for a RESTful service. Each of the dependencies and ports for each container is defined here: the REST service depends on gRPC, and the gRPC service depends on MongoDB. Each of the ports that these containers connect through is defined here as well. Executing this file (docker-compose) runs and implements all the services together with a single command in their respective containers.

This file is executed using the following commands:

```
docker-compose up --build -d      # Build the containers and run them
docker-compose logs -f            # This command is used to view all logs
docker-compose logs -f [service-name] # This command is used to view service-specific logs
```

Proto file modifications

CRUD functions to read, write, update, and delete items have been added for this task. This file is compiled to generate gRPC stubs that are used by the RESTful service to send, receive, updated and delete the communicated data accordingly.

gRPC server modifications

The main difference in control flow between the current task and the previous task is that MongoDB is actively involved between every read and write. When the REST service creates, updates or deletes an item, the gRPC stubs interact with MongoDB, which reserves a volume of storage upon inception. This is used to store the items created by the user that are sent via the REST service. Similarly, when the user requests a read operation, the data is read from MongoDB's database and provided to the user. The data is persistent, meaning that even if the containers cease to exist, the data can be accessed when the containers are restarted. However, Docker commands to prune the created volumes are also available. This will cause the stored data to be deleted. MongoDB is hosted on port 27017, and the gRPC service is hosted on port 50051.

Command to remove all different volume partitions reserved for the containers' databases:

```
docker system prune -a --volumes -f # Remove all volume partitions
```

RESTful service modifications

The major changes include the addition of circuit breakers in this script. Circuit breakers are configured for each method to open after 3 consecutive failures. The first try happens immediately, the second try happens after 100ms, and the third try happens after 200ms. After a reset timeout of 30 seconds, the connection status changed from open to half-open. This means that any new request will be accepted after 30 seconds, and the circuit will close again. The gRPC stubs generated by compiling the .proto file of the gRPC service is imported to the REST service as the stubs need to be accessed for data manipulation in MongoDB. This service is hosted on port 5000. A new route '/health' is added to check the functionality status of the circuit (closed/open/half-open).

Working

The Docker Compose file orchestrates the build and startup of all defined services. Individual service logs can be monitored to track the precise execution flow and identify issues.

Test Operations and Results

Build and start containers using a compose file

Containers are created and started using the following command:

`docker-compose up` # docker-compose is the name of the compose file

```
bavanmendon@Electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ docker-compose up --build -d
Creating network "compose-lab_app-network" with driver "bridge"
Creating volume "compose-lab_mongo-data" with default driver
Building grpc-service
[+] Building 1.2s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 330B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f75559
=> [internal] load build context
=> => transferring context: 424B
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt .
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> CACHED [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:61a267568f67bc4659ff15eb5a6e8637aeb277fce0bfcd560a508086b2b3bee1
=> => naming to docker.io/library/compose-lab_grpc-service
Building rest-service
[+] Building 0.3s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 200B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f75559
=> [internal] load build context
=> => transferring context: 7.24kB
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt .
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:5748fd16f795dfa586c7739cd0cbd699a095ac9cd05580a07da3e7714ba36068
=> => naming to docker.io/library/compose-lab_rest-service
Creating mongodb ... done
Creating grpc-service ... done
Creating rest-service ... done
```

List of containers created and started

```
bavannendon@Electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/compose-lab$ docker-compose ps
```

Name	Command	State	Ports
grpc-service	python3 -u server.py	Up	0.0.0.0:50051->50051/tcp,:::50051->50051/tcp
mongodb	docker-entrypoint.sh mongod	Up	0.0.0.0:27017->27017/tcp,:::27017->27017/tcp
rest-service	python3 app.py	Up	0.0.0.0:5000->5000/tcp,:::5000->5000/tcp

List of items created using CRUD commands via the REST service and their logs

Items are successfully created via POST requests to the REST service.

```
bavannendon@Electrofin:~$ curl -X POST http://localhost:5000/items -H "Content-Type: application/json" -d '{"id":1,"message":"Item created","name":"Kala"}'
bavannendon@Electrofin:~$ curl -X POST http://localhost:5000/items -H "Content-Type: application/json" -d '{"id":2,"name":"JP"}'
bavannendon@Electrofin:~$ curl -X POST http://localhost:5000/items -H "Content-Type: application/json" -d '{"id":3,"name":"Prapti"}'
bavannendon@Electrofin:~$ curl -X POST http://localhost:5000/items -H "Content-Type: application/json" -d '{"id":4,"name":"Zoey"}'
bavannendon@Electrofin:~$ curl -X POST http://localhost:5000/items -H "Content-Type: application/json" -d '{"id":5,"name":"Bavan"}'
```

```
grpc-service | [grpc] Creating Item: id=1, name=Kala
grpc-service | [grpc] Item created successfully: 1
rest-service | 172.18.0.1 - - [13/Feb/2026 11:52:02] "POST /items HTTP/1.1" 201 -
grpc-service | [grpc] Creating Item: id=2, name=JP
grpc-service | [grpc] Item created successfully: 2
rest-service | 172.18.0.1 - - [13/Feb/2026 11:52:17] "POST /items HTTP/1.1" 201 -
grpc-service | [grpc] Creating Item: id=3, name=Prapti
grpc-service | [grpc] Item created successfully: 3
rest-service | 172.18.0.1 - - [13/Feb/2026 11:52:25] "POST /items HTTP/1.1" 201 -
grpc-service | [grpc] Creating Item: id=4, name=Zoey
grpc-service | [grpc] Item created successfully: 4
rest-service | 172.18.0.1 - - [13/Feb/2026 11:52:32] "POST /items HTTP/1.1" 201 -
grpc-service | [grpc] Creating Item: id=5, name=Bavan
grpc-service | [grpc] Item created successfully: 5
rest-service | 172.18.0.1 - - [13/Feb/2026 11:52:40] "POST /items HTTP/1.1" 201 -
```

List of items read using the 'GET' operation, and its log

Items are retrieved using GET operations.

```
bavannendong@electrofin:~$ curl -X GET http://localhost:5000/items
[{"id":1,"name":"kala"}, {"id":2,"name":"JP"}, {"id":3,"name":"Prapti"}, {"id":4,"name":"Zoey"}, {"id":5,"name":"Bavan"}]
bavannendong@electrofin:~$

grpc-service | [gRPC] Creating Item: id=1, name=Kala
grpc-service | [gRPC] Item created successfully: 1
rest-service | 172.18.0.1 - - [13/Feb/2026 11:52:02] "POST /items HTTP/1.1" 201 -
grpc-service | [gRPC] Creating Item: id=2, name=JP
grpc-service | [gRPC] Item created successfully: 2
rest-service | 172.18.0.1 - - [13/Feb/2026 11:52:17] "POST /items HTTP/1.1" 201 -
grpc-service | [gRPC] Creating Item: id=3, name=Prapti
grpc-service | [gRPC] Item created successfully: 3
rest-service | 172.18.0.1 - - [13/Feb/2026 11:52:25] "POST /items HTTP/1.1" 201 -
grpc-service | [gRPC] Creating Item: id=4, name=Zoey
grpc-service | [gRPC] Item created successfully: 4
rest-service | 172.18.0.1 - - [13/Feb/2026 11:52:32] "POST /items HTTP/1.1" 201 -
grpc-service | [gRPC] Creating Item: id=5, name=Bavan
grpc-service | [gRPC] Item created successfully: 5
rest-service | 172.18.0.1 - - [13/Feb/2026 11:52:40] "POST /items HTTP/1.1" 201 -
bavannendong@electrofin:~$ curl -X GET http://localhost:5000/items
[{"id":1,"name":"kala"}, {"id":2,"name":"JP"}, {"id":3,"name":"Prapti"}, {"id":4,"name":"Zoey"}, {"id":5,"name":"Bavan"}]
bavannendong@electrofin:~$
```

Log of an item (ID: 5) updated

Item with ID 5 is successfully updated.

```
bavannendong@electrofin:~$ curl -X DELETE http://localhost:5000/items/5 -H "Content-Type: application/json" -d '{"id":5}'
{"id":5,"message":"Item deleted","name":"Bavan"}
bavannendong@electrofin:~$ curl -X PUT http://localhost:5000/items/5 -H "Content-Type: application/json" -d '{"id":5,"name":"Bavan Updated"}'
{"id":5,"message":"Item updated","name":"Bavan Updated"}
bavannendong@electrofin:~$ curl -X GET http://localhost:5000/items
[{"id":1,"name":"kala"}, {"id":2,"name":"JP"}, {"id":3,"name":"Prapti"}, {"id":4,"name":"Zoey"}, {"id":5,"name":"Bavan Updated"}]
bavannendong@electrofin:~$

grpc-service | [gRPC] Updating Item: id=5, name=Bavan Updated
grpc-service | [gRPC] Item updated successfully: 5
rest-service | 172.18.0.1 - - [13/Feb/2026 11:58:00] "PUT /items/5 HTTP/1.1" 201 -
grpc-service | [gRPC] Deleting Item: id=5, name=Bavan Updated
grpc-service | [gRPC] Item deleted successfully: 5
rest-service | 172.18.0.1 - - [13/Feb/2026 11:58:49] "DELETE /items/5 HTTP/1.1" 201 -
grpc-service | [gRPC] Creating Item: id=5, name=Bavan
grpc-service | [gRPC] Item created successfully: 5
rest-service | 172.18.0.1 - - [13/Feb/2026 11:59:34] "POST /items HTTP/1.1" 201 -
grpc-service | [gRPC] Updating Item: id=5, name=Bavan Updated
grpc-service | [gRPC] Item updated successfully: 5
rest-service | 172.18.0.1 - - [13/Feb/2026 11:59:47] "PUT /items/5 HTTP/1.1" 201 -
rest-service | 172.18.0.1 - - [13/Feb/2026 12:00:00] "GET /items HTTP/1.1" 200 -
```


Log of an item deleted (ID:5)

Item with ID 5 is successfully deleted.

```
bavannendon@electrofin:~$ curl -X DELETE http://localhost:5000/items/5 -H 'Content-Type: application/json' -d '{"id": 5}'
{"id":5,"message":"Item deleted","name":""}
bavannendon@electrofin:~$ curl -X GET http://localhost:5000/items
[{"id":1,"name":"Kala"}, {"id":2,"name":"JP"}, {"id":3,"name":"Praptti"}, {"id":4,"name":"Zoey"}]
bavannendon@electrofin:~$
```

```
rest-service | 172.18.0.1 - - [13/Feb/2026 11:54:55] "GET /items HTTP/1.1" 200 -
rest-service | 172.18.0.1 - - [13/Feb/2026 11:55:05] "GET /items HTTP/1.1" 200 -
mongodb | [{"ts":{"$date":"2026-02-13T11:55:33.621+00:00"},"s":"I", "c":"MONGODB", "id":22430, "ctx":{"Checkpointner","msg":"WiredTiger message","attr":{"message":{"ts_sec":1770983733,"ts_usec":621719,"thread":"110x7ab1572e56c0","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","category_id":7,"verbose_level":"DEBUG_1","verbose_level_id":1,"msg":"saving checkpoint snapshot mni: 8 5, snapshot max: 85 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1'}}}}]
mongodb | [{"ts":{"$date":"2026-02-13T11:56:33.637+00:00"},"s":"I", "c":"MONGODB", "id":22430, "ctx":{"Checkpointner","msg":"WiredTiger message","attr":{"message":{"ts_sec":1770983793,"ts_usec":637018,"thread":"110x7ab1572e56c0","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","category_id":7,"verbose_level":"DEBUG_1","verbose_level_id":1,"msg":"saving checkpoint snapshot mni: 8 5, snapshot max: 85 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1'}}}}]
mongodb | [{"ts":{"$date":"2026-02-13T11:57:33.653+00:00"},"s":"I", "c":"MONGODB", "id":22430, "ctx":{"Checkpointner","msg":"WiredTiger message","attr":{"message":{"ts_sec":1770983853,"ts_usec":653701,"thread":"110x7ab1572e56c0","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","category_id":7,"verbose_level":"DEBUG_1","verbose_level_id":1,"msg":"saving checkpoint snapshot mni: 8 5, snapshot max: 85 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1'}}}}]
grpc-service | [gRPC] Updating Item: id=5, name=Bavan Updated
rest-service | 172.18.0.1 - - [13/Feb/2026 11:58:00] "PUT /items/5 HTTP/1.1" 201 -
rest-service | 172.18.0.1 - - [13/Feb/2026 11:58:00] "DELETE /items/5 HTTP/1.1" 201 -
mongodb | [{"ts":{"$date":"2026-02-13T11:58:33.670+00:00"},"s":"I", "c":"MONGODB", "id":22430, "ctx":{"Checkpointner","msg":"WiredTiger message","attr":{"message":{"ts_sec":1770983913,"ts_usec":670497,"thread":"110x7ab1572e56c0","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","category_id":7,"verbose_level":"DEBUG_1","verbose_level_id":1,"msg":"saving checkpoint snapshot mni: 9 0, snapshot max: 90 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1'}}}}]
grpc-service | [gRPC] Deleting Item: id=5, name=
rest-service | 172.18.0.1 - - [13/Feb/2026 11:58:49] "DELETE /items/5 HTTP/1.1" 201 -
rest-service | 172.18.0.1 - - [13/Feb/2026 11:58:49] "DELETE /items/5 HTTP/1.1" 201 -
mongodb | [{"ts":{"$date":"2026-02-13T12:00:33.712+00:00"},"s":"I", "c":"MONGODB", "id":22430, "ctx":{"Checkpointner","msg":"WiredTiger message","attr":{"message":{"ts_sec":1770984033,"ts_usec":711922,"thread":"110x7ab1572e56c0","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","category_id":7,"verbose_level":"DEBUG_1","verbose_level_id":1,"msg":"saving checkpoint snapshot mni: 9 0, snapshot max: 90 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1'}}}}]
grpc-service | [gRPC] Creating Item: id=5, name=Bavan
rest-service | 172.18.0.1 - - [13/Feb/2026 11:59:34] "POST /items HTTP/1.1" 201 -
rest-service | 172.18.0.1 - - [13/Feb/2026 11:59:34] "POST /items HTTP/1.1" 201 -
grpc-service | [gRPC] Updating Item: id=5, name=Bavan Updated
rest-service | 172.18.0.1 - - [13/Feb/2026 11:59:47] "PUT /items/5 HTTP/1.1" 201 -
rest-service | 172.18.0.1 - - [13/Feb/2026 12:00:00] "GET /items HTTP/1.1" 200 -
grpc-service | [gRPC] Deleting Item: id=5, name=
rest-service | 172.18.0.1 - - [13/Feb/2026 12:00:32] "DELETE /items/5 HTTP/1.1" 201 -
rest-service | 172.18.0.1 - - [13/Feb/2026 12:00:32] "DELETE /items/5 HTTP/1.1" 201 -
mongodb | [{"ts":{"$date":"2026-02-13T12:00:33.712+00:00"},"s":"I", "c":"MONGODB", "id":22430, "ctx":{"Checkpointner","msg":"WiredTiger message","attr":{"message":{"ts_sec":1770984033,"ts_usec":711922,"thread":"110x7ab1572e56c0","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","category_id":7,"verbose_level":"DEBUG_1","verbose_level_id":1,"msg":"saving checkpoint snapshot mni: 9 0, snapshot max: 90 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1'}}}}]
rest-service | 172.18.0.1 - - [13/Feb/2026 12:00:34] "GET /items HTTP/1.1" 200 -
```

Checking the health of the connection with logs

Circuit breaker status is monitored via the '/health' endpoint.

```
bavannendon@electrofin:~$ curl -X GET http://localhost:5000/health
{"circuit_breaker_state":"closed","status":"healthy"}
bavannendon@electrofin:~$
```

```
4, snapshot max: 84 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1'}}}}]
mongodb | [{"ts":{"$date":"2026-02-13T11:56:33.637+00:00"},"s":"I", "c":"MONGODB", "id":22430, "ctx":{"Checkpointner","msg":"WiredTiger message","attr":{"message":{"ts_sec":1770983733,"ts_usec":621719,"thread":"110x7ab1572e56c0","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","category_id":7,"verbose_level":"DEBUG_1","verbose_level_id":1,"msg":"saving checkpoint snapshot mni: 8 5, snapshot max: 85 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1'}}}}]
mongodb | [{"ts":{"$date":"2026-02-13T11:57:33.653+00:00"},"s":"I", "c":"MONGODB", "id":22430, "ctx":{"Checkpointner","msg":"WiredTiger message","attr":{"message":{"ts_sec":1770983853,"ts_usec":653701,"thread":"110x7ab1572e56c0","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","category_id":7,"verbose_level":"DEBUG_1","verbose_level_id":1,"msg":"saving checkpoint snapshot mni: 8 5, snapshot max: 85 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1'}}}}]
mongodb | [{"ts":{"$date":"2026-02-13T11:57:33.653+00:00"},"s":"I", "c":"MONGODB", "id":22430, "ctx":{"Checkpointner","msg":"WiredTiger message","attr":{"message":{"ts_sec":1770983853,"ts_usec":653701,"thread":"110x7ab1572e56c0","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","category_id":7,"verbose_level":"DEBUG_1","verbose_level_id":1,"msg":"saving checkpoint snapshot mni: 8 5, snapshot max: 85 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1'}}}}]
grpc-service | [gRPC] Updating Item: id=5, name=Bavan Updated
rest-service | 172.18.0.1 - - [13/Feb/2026 11:58:00] "PUT /items/5 HTTP/1.1" 201 -
rest-service | 172.18.0.1 - - [13/Feb/2026 11:58:00] "DELETE /items/5 HTTP/1.1" 201 -
mongodb | [{"ts":{"$date":"2026-02-13T11:58:33.670+00:00"},"s":"I", "c":"MONGODB", "id":22430, "ctx":{"Checkpointner","msg":"WiredTiger message","attr":{"message":{"ts_sec":1770983913,"ts_usec":670497,"thread":"110x7ab1572e56c0","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","category_id":7,"verbose_level":"DEBUG_1","verbose_level_id":1,"msg":"saving checkpoint snapshot mni: 9 0, snapshot max: 90 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1'}}}}]
grpc-service | [gRPC] Deleting Item: id=5, name=
rest-service | 172.18.0.1 - - [13/Feb/2026 11:58:49] "DELETE /items/5 HTTP/1.1" 201 -
rest-service | 172.18.0.1 - - [13/Feb/2026 11:58:49] "DELETE /items/5 HTTP/1.1" 201 -
mongodb | [{"ts":{"$date":"2026-02-13T12:00:33.712+00:00"},"s":"I", "c":"MONGODB", "id":22430, "ctx":{"Checkpointner","msg":"WiredTiger message","attr":{"message":{"ts_sec":1770984033,"ts_usec":711922,"thread":"110x7ab1572e56c0","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","category_id":7,"verbose_level":"DEBUG_1","verbose_level_id":1,"msg":"saving checkpoint snapshot mni: 9 0, snapshot max: 90 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1'}}}}]
grpc-service | [gRPC] Creating Item: id=5, name=Bavan
rest-service | 172.18.0.1 - - [13/Feb/2026 11:59:34] "POST /items HTTP/1.1" 201 -
rest-service | 172.18.0.1 - - [13/Feb/2026 11:59:34] "POST /items HTTP/1.1" 201 -
grpc-service | [gRPC] Updating Item: id=5, name=Bavan Updated
rest-service | 172.18.0.1 - - [13/Feb/2026 11:59:47] "PUT /items/5 HTTP/1.1" 201 -
rest-service | 172.18.0.1 - - [13/Feb/2026 12:00:00] "GET /items HTTP/1.1" 200 -
grpc-service | [gRPC] Deleting Item: id=5, name=
rest-service | 172.18.0.1 - - [13/Feb/2026 12:00:32] "DELETE /items/5 HTTP/1.1" 201 -
rest-service | 172.18.0.1 - - [13/Feb/2026 12:00:32] "DELETE /items/5 HTTP/1.1" 201 -
mongodb | [{"ts":{"$date":"2026-02-13T12:00:33.712+00:00"},"s":"I", "c":"MONGODB", "id":22430, "ctx":{"Checkpointner","msg":"WiredTiger message","attr":{"message":{"ts_sec":1770984033,"ts_usec":711922,"thread":"110x7ab1572e56c0","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","category_id":7,"verbose_level":"DEBUG_1","verbose_level_id":1,"msg":"saving checkpoint snapshot mni: 9 0, snapshot max: 90 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1'}}}}]
rest-service | 172.18.0.1 - - [13/Feb/2026 12:00:34] "GET /items HTTP/1.1" 200 -
rest-service | 172.18.0.1 - - [13/Feb/2026 12:01:34] "GET /health HTTP/1.1" 200 -
```

Failure Scenario Testing

GRPC Service Failure (Circuit Breaker Test)

Terminating the gRPC service triggers the retry mechanism. The request is retried three times before failing with a 504 status code, indicating that the server is unavailable.

```
bavannendon@electrofin:~$ cd Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/
bavannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals$ cd
compose-lab/
bavannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ ls
docker-compose.yml  grpc-service  rest-service
bavannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ docker-compose stop grpc-service/
ERROR: No such service: grpc-service/
bavannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ docker-compose stop grpc-service
Stopping grpc-service ... done
bavannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ curl -X POST http://localhost:5000/items -H "Content-Type: application/json" -d '{"id": 0, "nam
e": "Test failure"}'
{"error": "Service unavailable"}
bavannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$
```

```
thread": "10x7ab1572e56c0", "session_name": "WT_SESSION_checkpoint", "category": "WT_VERB_CHECKPOINT_PROGRESS",
"category_id": 7, "verbose_level": "DEBUG_1", "verbose_level_id": 1, "msg": "saving checkpoint snapshot mni: 2
09, snapshot max: 209 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) bas
e write gen: 1"]]]
mongodb | [{"ts":{"sdate":"2026-02-13T13:49:35.729+00:00"},"s":"I", "c":"WTCHKPT", "id":"22430,
"ctx":{"checkpointnr": "msg":"WiredTiger message", "attr":{"message":{"ts_sec":1770990575,"ts_usec":729708,"
thread":"10x7ab1572e56c0","session_name":"WT_SESSION_checkpoint", "category":"WT_VERB_CHECKPOINT_PROGRESS",
"category_id":7, "verbose_level":"DEBUG_1", "verbose_level_id":1, "msg":"saving checkpoint snapshot mni: 2
10, snapshot max: 210 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) bas
e write gen: 1"]]]
mongodb | [{"ts":{"sdate":"2026-02-13T13:50:35.748+00:00"},"s":"I", "c":"WTCHKPT", "id":"22430,
"ctx":{"checkpointnr": "msg":"WiredTiger message", "attr":{"message":{"ts_sec":1770990635,"ts_usec":748269,"
thread":"10x7ab1572e56c0","session_name":"WT_SESSION_checkpoint", "category":"WT_VERB_CHECKPOINT_PROGRESS",
"category_id":7, "verbose_level":"DEBUG_1", "verbose_level_id":1, "msg":"saving checkpoint snapshot mni: 2
11, snapshot max: 211 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) bas
e write gen: 1"]]]
mongodb | [{"ts":{"sdate":"2026-02-13T13:51:35.765+00:00"},"s":"I", "c":"WTCHKPT", "id":"22430,
"ctx":{"checkpointnr": "msg":"WiredTiger message", "attr":{"message":{"ts_sec":1770990695,"ts_usec":765444,"
thread":"10x7ab1572e56c0","session_name":"WT_SESSION_checkpoint", "category":"WT_VERB_CHECKPOINT_PROGRESS",
"category_id":7, "verbose_level":"DEBUG_1", "verbose_level_id":1, "msg":"saving checkpoint snapshot mni: 2
12, snapshot max: 212 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) bas
e write gen: 1"]]]
mongodb | [{"ts":{"sdate":"2026-02-13T13:52:35.782+00:00"},"s":"I", "c":"WTCHKPT", "id":"22430,
"ctx":{"checkpointnr": "msg":"WiredTiger message", "attr":{"message":{"ts_sec":1770990755,"ts_usec":782576,"
thread":"10x7ab1572e56c0","session_name":"WT_SESSION_checkpoint", "category":"WT_VERB_CHECKPOINT_PROGRESS",
"category_id":7, "verbose_level":"DEBUG_1", "verbose_level_id":1, "msg":"saving checkpoint snapshot mni: 2
13, snapshot max: 213 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) bas
e write gen: 1"]]]
Exception in thread Thread-4:
Traceback (most recent call last):
  File "threading.py", line 926, in _bootstrap_inner
  File "threading.py", line 870, in run
  File "compose/cli/log_printer.py", line 202, in watch_events
  File "compose/project.py", line 626, in yield_loop
  File "compose/project.py", line 594, in build_container_event
KeyError: 'id'
mongodb | [{"ts":{"sdate":"2026-02-13T13:53:07.374+00:00"},"s":"I", "c":"-", "id":"20883,
"ctx":{"conn1": "msg":"Interrupted operation as its client disconnected", "attr":{"opid":"12047"}
mongodb | [{"ts":{"sdate":"2026-02-13T13:53:07.374+00:00"},"s":"I", "c":"NETWORK", "id":"22944,
"ctx":{"conn3": "msg":"Connection ended", "attr":{"remote":"172.18.0.3:35500","loadBalanced":false,"uid":
("uid":{"suid":"dfdc2d7-2afa-486c-acf9-77be28cad930"},"connectionId":3,"connectionCount":12)}
mongodb | [{"ts":{"sdate":"2026-02-13T13:53:07.374+00:00"},"s":"I", "c":"NETWORK", "id":"22944,
"ctx":{"conn2": "msg":"Connection ended", "attr":{"remote":"172.18.0.3:35088","loadBalanced":false,"uid":
("uid":{"suid":"17a92fde4-a50d-404e-9053-9dc2d8a13595"},"connectionId":2,"connectionCount":11)}
mongodb | [{"ts":{"sdate":"2026-02-13T13:53:07.375+00:00"},"s":"I", "c":"NETWORK", "id":"22944,
"ctx":{"conn1": "msg":"Connection ended", "attr":{"remote":"172.18.0.3:35082","loadBalanced":false,"uid":
("uid":{"suid":"04601cdc-acf7-4211-b04e-0dfa57e1a78f"},"connectionId":1,"connectionCount":0)}
grpc-service exited with code 137
rest-service | 172.18.0.1 -- [13/Feb/2026 13:53:29] "POST /items HTTP/1.1" 503 -
mongodb | [{"ts":{"sdate":"2026-02-13T13:53:35.799+00:00"},"s":"I", "c":"WTCHKPT", "id":"22430,
"ctx":{"checkpointnr": "msg":"WiredTiger message", "attr":{"message":{"ts_sec":1770990815,"ts_usec":820669,"
thread":"10x7ab1572e56c0","session_name":"WT_SESSION_checkpoint", "category":"WT_VERB_CHECKPOINT_PROGRESS",
"category_id":7, "verbose_level":"DEBUG_1", "verbose_level_id":1, "msg":"saving checkpoint snapshot mni: 2
14, snapshot max: 214 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) bas
e write gen: 1"]]]
[]
```

MongoDB Failure (Database Failure Test)

Killing the MongoDB service produces a similar response.

```
bavannendon@electrofin:~$ cd Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/
bavannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals$ cd
compose-lab/
bavannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ ls
docker-compose.yml  grpc-service  rest-service
bavannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ docker-compose stop grpc-service/
ERROR: No such service: grpc-service/
bavannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ docker-compose stop grpc-service
Stopping grpc-service ... done
bavannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ curl -X POST http://localhost:5000/items -H "Content-Type: application/json" -d '{"id": 0, "nam
e": "Test failure"}'
{"error": "Service unavailable"}
bavannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ docker-compose restart grpc-service
Restarting grpc-service ... done
bavannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ docker-compose stop mongod
Stopping mongod ... done
bavannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ curl -X POST http://localhost:5000/items -H "Content-Type: application/json" -d '{"id": 0, "nam
e": "Test failure"}'
{"error": "Service unavailable"}
bavannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$
```

```
mongodb | [{"ts":{"sdate":"2026-02-13T13:57:11.819+00:00"},"s":"I", "c":"NETWORK", "id":"22944,
"ctx":{"conn5": "msg":"Connection ended", "attr":{"remote":"172.18.0.3:35192","loadBalanced":false,"uid":
("uid":{"suid":"2206f3c0-3eef-41d0-a61c-50a954a2802c"},"connectionId":5,"connectionCount":11)}
mongodb | [{"ts":{"sdate":"2026-02-13T13:57:11.819+00:00"},"s":"I", "c":"STORAGE", "id":"22317,
"ctx":{"signalHandler": "msg":"WiredTigerEngine shutting down"}
mongodb | [{"ts":{"sdate":"2026-02-13T13:57:11.819+00:00"},"s":"I", "c":"STORAGE", "id":"22318,
"ctx":{"signalHandler": "msg":"Shutting down session sweeper thread"}
mongodb | [{"ts":{"sdate":"2026-02-13T13:57:11.819+00:00"},"s":"I", "c":"STORAGE", "id":"22319,
"ctx":{"signalHandler": "msg":"Finished shutting down session sweeper thread"}
mongodb | [{"ts":{"sdate":"2026-02-13T13:57:11.819+00:00"},"s":"I", "c":"STORAGE", "id":"4795902,
"ctx":{"signalHandler": "msg":"Closing WiredTiger", "attr":{"closeConfig":{"leak_memory=true,use_timestampfa
lse,"}}
mongodb | [{"ts":{"sdate":"2026-02-13T13:57:11.820+00:00"},"s":"I", "c":"WTCHKPT", "id":"22430,
"ctx":{"signalHandler": "msg":"WiredTiger message", "attr":{"message":{"ts_sec":1770991031,"ts_usec":820669,"
thread":"10x7ab15b026c0","session_name":"close ckpt", "category":"WT_VERB_CHECKPOINT_PROGRESS", "category_
y_id":7, "verbose_level":"DEBUG_1", "verbose_level_id":1, "msg":"saving checkpoint snapshot mni: 218, snapsho
t max: 218 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write ge
n: 1"]]]
mongodb | [{"ts":{"sdate":"2026-02-13T13:57:11.823+00:00"},"s":"I", "c":"WTRECOV", "id":"22430,
"ctx":{"signalHandler": "msg":"WiredTiger message", "attr":{"message":{"ts_sec":1770991031,"ts_usec":823744,"
thread":"10x7ab15b026c0","session_name":"WT_CONNECTION_close", "category":"WT_VERB_RECOVERY_PROGRESS",
"category_id":134, "verbose_level":"DEBUG_1", "verbose_level_id":1, "msg":"shutdown checkpoint has successfull
y finished and ran for 3 milliseconds"]]]
mongodb | [{"ts":{"sdate":"2026-02-13T13:57:11.823+00:00"},"s":"I", "c":"WTRECOV", "id":"22430,
"ctx":{"signalHandler": "msg":"WiredTiger message", "attr":{"message":{"ts_sec":1770991031,"ts_usec":823849,"
thread":"10x7ab15b026c0","session_name":"WT_CONNECTION_close", "category":"WT_VERB_RECOVERY_PROGRESS",
"category_id":134, "verbose_level":"DEBUG_1", "verbose_level_id":1, "msg":"shutdown was completed successfully
and took 3ms, including 0ms for the rollback to stable, and 3ms for the checkpoint."}}
mongodb | [{"ts":{"sdate":"2026-02-13T13:57:11.830+00:00"},"s":"I", "c":"STORAGE", "id":"4795901,
"ctx":{"signalHandler": "msg":"WiredTiger closed", "attr":{"durationInMills":133}
mongodb | [{"ts":{"sdate":"2026-02-13T13:57:11.830+00:00"},"s":"I", "c":"STORAGE", "id":"22279,
"ctx":{"signalHandler": "msg":"shutdown: removing fs lock..."}
mongodb | [{"ts":{"sdate":"2026-02-13T13:57:11.830+00:00"},"s":"I", "c":"-", "id":"4784931,
"ctx":{"signalHandler": "msg":"Dropping the scope cache for shutdown"}
mongodb | [{"ts":{"sdate":"2026-02-13T13:57:11.830+00:00"},"s":"I", "c":"FTDC", "id":"20626,
"ctx":{"signalHandler": "msg":"Shutting down full-time diagnostic data capture"}
mongodb | [{"ts":{"sdate":"2026-02-13T13:57:11.837+00:00"},"s":"I", "c":"CONTROL", "id":"20565,
"ctx":{"signalHandler": "msg":"Now exiting"}
mongodb | [{"ts":{"sdate":"2026-02-13T13:57:11.837+00:00"},"s":"I", "c":"CONTROL", "id":"8423404,
"ctx":{"signalHandler": "msg":"mongod shutdown complete", "attr":{"Summary of time elapsed":{"Enter terminal shutdown":"0 ms","Step down the replication coordinator for shutdown":"0 ms","Time spent i
n queue mode":"0 ms","Shut down FILE Crud subsystem":"0 ms","Shut down WiredTiger recovery":"0 ms","Shut down
WaitForMajorityService":"1 ms","Shut down the logical session cache":"0 ms","Shut down the global connectio
n pool":"0 ms","Shut down the flow control ticket holder":"0 ms","Shut down the thread that aborts expir
ed transactions":"0 ms","Kill all operations for shutdown":"0 ms","Shut down all tenant migration access
blockers on global shutdown":"0 ms","Shut down all open transactions":"0 ms","Acquire the ASTL for shutd
own":"0 ms","Shut down the IndexBuildCoordinator and wait for index builds to finish":"0 ms","Shut down
the replica set monitor":"0 ms","Shut down the migration util executor":"1 ms","Shut down the transport l
ayer":"0 ms","Shut down the health log":"0 ms","Shut down the TTL monitor":"0 ms","Shut down expired pre-
pares and documents removers":"0 ms","Shut down the storage engine":"11 ms","Wait for the oplog cap main
rainer thread to stop":"0 ms","Shut down full-time data capture":"7 ms","Shut down online certificate sta
tus protocol manager":"0 ms","shutdownTask total elapsed time":"20 ms"]]]
mongodb | [{"ts":{"sdate":"2026-02-13T13:57:11.837+00:00"},"s":"I", "c":"CONTROL", "id":"23138,
"ctx":{"signalHandler": "msg":"Shutting down", "attr":{"exitcode":0}
mongodb exited with code 0
rest-service | 172.18.0.1 -- [13/Feb/2026 13:57:22] "POST /items HTTP/1.1" 503 -
[]
```


Verification through MongoDB shell access after service restart confirms that data sent during the outage was not persisted. However, previously stored data remains intact, demonstrating proper persistence functionality.

```
pose-labs curl -X POST http://localhost:5000/items -H "Content-Type: application/json" -d '{"id": 6, "name": "Test failure"}'
{"error": "Service unavailable"}
havanonnend@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/compose-labs$ docker-compose restart grpc-service
Restarting grpc-service ... done
havanonnend@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/compose-labs$ docker-compose stop mongodb
Stopping mongodb ... done
havanonnend@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/compose-labs$ curl -X POST http://localhost:5000/items -H "Content-Type: application/json" -d '{"id": 6, "name": "Test failure"}'
{"error": "Service unavailable"}
havanonnend@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/compose-labs$ docker-compose start mongodb
Starting mongodb ... done
havanonnend@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/compose-labs$ docker exec it_mongodb mongosh
Current Mongosh Log ID: 699f2e42c64919243bbaa08
Connecting to:
  mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.5.3
Using MongoDB:
  2.5.3
Using Mongosh:
```

For mongosh info see: <https://www.mongodb.com/docs/mongodb-shell/>

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (<https://www.mongodb.com/legal/privacy-policy>).
You can opt-out by running the `disableTelemetry()` command.

```

The server generated these startup warnings when booting
2026-02-13T13:59:19.441Z:000: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See http://doc.mongodb.org/core/prodnotes-filesystem
2026-02-13T13:59:19.441+0000: Access control is not enabled for the database. Read and write access to
data and configuration is unrestricted
2026-02-13T13:59:19.441+0000: Soft limits for open file descriptors too low
2026-02-13T13:59:19.441+0000: For customers running the current memory allocator, we suggest changing
the contents of the following sysfsfile
sys-02-13T13:59:19.441+0000: For customers running the current memory allocator, we suggest changing
the contents of the following sysfsfile
2026-02-13T13:59:19.441+0000: We suggest setting the contents of sysfsFile to 0
2026-02-13T13:59:19.441+0000: We suggest setting swappiness to 0 or 1, as swapping can cause performance
problems

```

```
jest> use itemsdb
switched to db itemsdb
itemsdb> db.items.find()

[ { _id: ObjectId('699f10627b754f8f665ea710'), id: 1, name: 'Kala' },
  { _id: ObjectId('699f10717b754f8f665ea711'), id: 2, name: '3P' },
  { _id: ObjectId('699f10797b754f8f665ea712'), id: 3, name: 'Praptti' },
  { _id: ObjectId('699f10807b754f8f665ea713'), id: 4, name: 'Zoey' } ]

itemsdb>
```

```

nongdb | ("t":{"$date":"2026-02-13T13:57:11.819+00:00"},"s":{"_id":"NETWORK","id":"22940",
"ctx":{"conn5","nsp":"Connection ended","attr":{"remote":"172.18.0.3:35192","isLoadBalanced":false,"uid":"",
"uid":{"$uid":"2206f3ce-3ee4-41d0-ae5c-589494a2862c"},"connectionId":"5","connectionContext":1}}},
nongdb | ("t":{"$date":"2026-02-02-13:57:11.819+00:00"},"s":{"t","c":"STORAGE","id":"22319",
"ctx":{"$signaHandler","nsp":"Shutting down session sweeper thread"}},
nongdb | ("t":{"$date":"2026-02-13:57:11.819+00:00"},"s":{"t","c":"STORAGE","id":"22318",
"ctx":{"$signaHandler","nsp":"Shutting down session sweeper thread"}},
nongdb | ("t":{"$date":"2026-02-13:57:11.819+00:00"},"s":{"t","c":"STORAGE","id":"22319",
"ctx":{"$signaHandler","nsp":"Shutting down session sweeper thread"}},
nongdb | ("t":{"$date":"2026-02-13:57:11.819+00:00"},"s":{"t","c":"STORAGE","id":"4795902",
"ctx":{"$signaHandler","nsp":"Closing WiredTiger","attr":{"closeConfig":{"leak_memory":true,use_testnames":fa
lse,"}}}},
nongdb | ("t":{"$date":"2026-02-13:57:11.820+00:00"},"s":{"t","c":"WTCMV","id":"22430",
"ctx":{"$signaHandler","nsp":"WiredTiger message","attr":{"message":{"ts_sec":"170991031","ts_usec":"820669",
"thread":"1:0x7ab165b026c0","session_name":"close_kcpt","category":"WT_VERB_CONNECTION_PROGRESS","category.i
d":"17","verbose_level":"DEBUG","verbose_level_id":"1","nsp":"saving checkpoint snapshot min: 218, snapsho
t max: 218 snapshot count: 0, oldest timestamp: (0, 0) , meta checkpoint timestamp: (0, 0) base write file
ise,"}}}},
nongdb | ("t":{"$date":"2026-02-13:57:11.823+00:00"},"s":{"t","c":"WTRECOVER","id":"22430",
"ctx":{"$signaHandler","nsp":"WiredTiger message","attr":{"message":{"ts_sec":"170991031","ts_usec":"823744",
"thread":"1:0x7ab165b026c0","session_name":"WT_CONNECTION.close","category":"WT_VERB_RECOVERY_PROGRESS","c
ategory.id":"18","verbose_level":"DEBUG","verbose_level_id":"1","nsp":"Shutdown checkpoint has successful
ly finished and ran for 0 milliseconds"}}}},
nongdb | ("t":{"$date":"2026-02-13:57:11.823+00:00"},"s":{"t","c":"WTRECOVER","id":"22430",
"ctx":{"$signaHandler","nsp":"WiredTiger message","attr":{"message":{"ts_sec":"170991031","ts_usec":"823849",
"thread":"1:0x7ab165b026c0","session_name":"WT_CONNECTION.close","category":"WT_VERB_RECOVERY_PROGRESS","c
ategory.id":"18","verbose_level":"DEBUG","verbose_level_id":"1","nsp":"Shutdown checkpoint has successful
ly finished and ran for 0 milliseconds and took 3ms, including bcs for the rollback to stable, and 3ms for the checkpoint.}}}},
nongdb | ("t":{"$date":"2026-02-13:57:11.830+00:00"},"s":{"t","c":"STORAGE","id":"4795901",
"ctx":{"$signaHandler","nsp":"WiredTiger closed","attr":{"durationMillis":11}}},
nongdb | ("t":{"$date":"2026-02-13:57:11.830+00:00"},"s":{"t","c":"STORAGE","id":"22279",
"ctx":{"$signaHandler","nsp":"shutdown: removing fs lock..."}},
nongdb | ("t":{"$date":"2026-02-13:57:11.830+00:00"},"s":{"t","c":"T","id":"7484931",
"ctx":{"$signaHandler","nsp":"Dropping the scope cache for shutdown"}},
nongdb | ("t":{"$date":"2026-02-13:57:11.837+00:00"},"s":{"t","c":"TDOC","id":"26026",
"ctx":{"$signaHandler","nsp":"Shutting down full time diagnostic data capture"}},
nongdb | ("t":{"$date":"2026-02-13:57:11.837+00:00"},"s":{"t","c":"CONTROL","id":"20565",
"ctx":{"$signaHandler","nsp":"Now exiting"}},
nongdb | ("t":{"$date":"2026-02-13:57:11.837+00:00"},"s":{"t","c":"CONTROL","id":"842340",
"ctx":{"$signaHandler","nsp":"Mongo shutdown complete","attr":{"Summary of time elapsed":{"Statistics":{"E
nvironmental shutdown":"0 ms","Step down the replication coordinator for shutdown":"0 ms","Time spent i
n queue mode":"0 ms","Shut down FLE CRUD subsystem":"0 ms","Shut down MirrorMaoestro":"0 ms","Shut down
WaitForMajorityService":"0 ms","Shut down the logical session cache":"0 ms","Shut down the global connect
ion cache":"0 ms","Shut down the global connection pool":"0 ms","Shut down the shard and the sharding op
erated transactions":"0 ms","Kill all operations for shutdown":"0 ms","Shut down all tenant migration ac
cess blockers on global shutdown":"0 ms","Shut down all open transactions":"0 ms","Acquire the RSTL for shut
down":"0 ms","Shut down the IndexBuildsCoordinator and wait for index builds to finish":"0 ms","Shut down
the meta set manager":"0 ms","Shut down the metadata manager":"0 ms","Shut down the transport l
ayer":"0 ms","Shut down the health log":"0 ms","Shut down the TTL monitor":"0 ms","Shut down expired pre
images and documents removers":"0 ms","Shut down the storage engine":"11 ms","Wait for the oplog cap main
tainer thread to stop":"0 ms","Shut down full-time data capture":"7 ms","Shut down online certificate sta
tus protocol manager":"0 ms","Shut down the oplog capped collection":"0 ms"},"}}}},
nongdb | ("t":{"$date":"2026-02-13:57:11.847+00:00"},"s":{"t","c":"CONTROL","id":"22138",
"ctx":{"$signaHandler","nsp":"Shutting down","attr":{"exitCode":0}}},
nongdb exited with code 0
rest-service 172.18.0.1 - [13/Feb/2026 13:57:22] "POST /status HTTP/1.1" 500 -

```

Volume Removal Test

Testing persistence after executing 'docker-compose down -v' verifies that volume removal permanently deletes all stored data.

```
Building rest-service:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals$ docker-compose up
[+] Building 0.2s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 200B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 0.2s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f75559 0.0s
=> [internal] load build context                                  0.0s
=> => transferring context: 320B                                      0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> CACHED [3/5] COPY requirements.txt .                             0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt 0.0s
=> CACHED [5/5] COPY . .                                           0.0s
=> exporting to image                                              0.0s
=> => exporting layers                                              0.0s
=> => writing image sha256:5748fd16f795dfa586c7739cd0cbd699a095ac9cd05580a07da3e7714ba36068 0.0s
=> => naming to docker.io/library/compose-lab_rest-service        0.0s
Creating mongodb ... done
Creating grpc-service ... done
Creating rest-service ... done
bavanmendon@Electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ docker exec -it mongodb mongosh
Current Mongosh Log ID: 698f2f23469894651bbaa8b8
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&ap
pName=mongosh+2.5.3
Using MongoDB:          8.0.11
Using Mongosh:          2.5.3

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

-----
The server generated these startup warnings when booting
2026-02-13T14:02:57.677+00:00: Using the XFS filesystem is strongly recommended with the WiredTiger st
orage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem
2026-02-13T14:02:58.223+00:00: Access control is not enabled for the database. Read and write access t
o data and configuration is unrestricted
2026-02-13T14:02:58.223+00:00: Soft rlimits for open file descriptors too low
2026-02-13T14:02:58.223+00:00: For customers running the current memory allocator, we suggest changing
the contents of the following sysfsFile
2026-02-13T14:02:58.223+00:00: For customers running the current memory allocator, we suggest changing
the contents of the following sysfsFile
2026-02-13T14:02:58.223+00:00: We suggest setting the contents of sysfsFile to 0.
2026-02-13T14:02:58.223+00:00: We suggest setting swappiness to 0 or 1, as swapping can cause performa
nce problems.
-----

test> use itemsdb
switched to db itemsdb
itemsdb> db.items.find()

itemsdb> █
```


Circuit Breaker Reset Timeout Test

This test validates the 30-second reset timeout mechanism:

- An item “Test 1” is successfully sent and stored.
- Reposting the same data triggers an error and opens the circuit, verifiable through the circuit breaker state endpoint.
- Attempting to send a second item fails because the circuit remains open.
- After 30 seconds, the circuit breaker transitions to half-open state.
- The second item can now be successfully transmitted.

```
havannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ curl -X POST http://localhost:5000/items -H "Content-Type: application/json" -d '{"id": 1, "name": "Test 1"}'
{"id":1,"message":"Item created","name":"Test 1"}
havannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ curl -X POST http://localhost:5000/items -H "Content-Type: application/json" -d '{"id": 1, "name": "Test 1"}'
{"error":"Service unavailable"}
havannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ curl -X POST http://localhost:5000/items -H "Content-Type: application/json" -d '{"id": 2, "name": "Test 2"}'
{"error":"Service unavailable"}
havannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ curl -X GET http://localhost:5000/health
{"circuit_breaker_state":"open","status":"healthy"}
havannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ curl -X POST http://localhost:5000/items -H "Content-Type: application/json" -d '{"id": 2, "name": "Test 2"}'
{"id":2,"message":"Item created","name":"Test 2"}
havannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$ curl -X GET http://localhost:5000/health
{"circuit_breaker_state":"closed","status":"healthy"}
havannendon@electrofin:~/Documents/Anhalt/distributed-software-architecture/practicals/dsa-practicals/com
pose-lab$
```

```
write gen: 1'))
rest-service | [REST] Connected to gRPC at grpc-service:50051
rest-service | * Serving Flask app 'app'
rest-service | * Debug mode: off
rest-service | WARNING: This is a development server. Do not use it in a production deployment. Use a
production WSGI server instead.
rest-service | * Running on all addresses (0.0.0.0)
rest-service | * Running on http://127.0.0.1:5000
rest-service | Press CTRL+C to quit
grpc-service | [gRPC] Creating item: id=1, name=Test 1
grpc-service | [gRPC] Item created successfully: 1
rest-service | 172.18.0.1 - - [13/Feb/2026 14:05:25] "POST /items HTTP/1.1" 201 -
grpc-service | [gRPC] Creating item: id=1, name=Test 1
grpc-service | [gRPC] Error creating item: E11000 duplicate key error collection: itemsdb.items index:
id 1 dup key: { id: 1 }, full error: { 'index': 0, 'code': 11000, 'errmsg': 'E11000 duplicate key error c
ollection: itemsdb.items index: id 1 dup key: { id: 1 }', 'keyPattern': { 'id': 1 }, 'keyValue': { 'id': 1 } }
grpc-service | [gRPC] Creating item: id=1, name=Test 1
grpc-service | [gRPC] Error creating item: E11000 duplicate key error collection: itemsdb.items index:
id 1 dup key: { id: 1 }, full error: { 'index': 0, 'code': 11000, 'errmsg': 'E11000 duplicate key error c
ollection: itemsdb.items index: id 1 dup key: { id: 1 }', 'keyPattern': { 'id': 1 }, 'keyValue': { 'id': 1 } }
grpc-service | [gRPC] Creating item: id=1, name=Test 1
grpc-service | [gRPC] Error creating item: E11000 duplicate key error collection: itemsdb.items index:
id 1 dup key: { id: 1 }, full error: { 'index': 0, 'code': 11000, 'errmsg': 'E11000 duplicate key error c
ollection: itemsdb.items index: id 1 dup key: { id: 1 }', 'keyPattern': { 'id': 1 }, 'keyValue': { 'id': 1 } }
rest-service | 172.18.0.1 - - [13/Feb/2026 14:05:35] "POST /items HTTP/1.1" 503 -
rest-service | 172.18.0.1 - - [13/Feb/2026 14:05:48] "POST /items HTTP/1.1" 503 -
mongod | [{"t":{"$date":"2026-02-13T14:05:50.291+00:00"},"s":"I", "c":"WTCHKPT", "id":22430,
"ctx":{"CheckpointInt","msg":{"WireTiger message","attr":{"message":{"ts_sec":1770991618,"ts_usec":273161,"
thread":"1:0x7b17921306c0","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS
","category_id":7,"verbose_level":"DEBUG_1","verbose_level_id":1,"msg":"saving checkpoint snapshot nIn: 5
2, snapshot max: 52 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base
write gen: 1'}}}}
rest-service | 172.18.0.1 - - [13/Feb/2026 14:06:25] "GET /health HTTP/1.1" 200 -
mongod | [{"t":{"$date":"2026-02-13T14:07:58.291+00:00"},"s":"I", "c":"WTCHKPT", "id":22430,
"ctx":{"CheckpointInt","msg":{"WireTiger message","attr":{"message":{"ts_sec":1770991678,"ts_usec":291772,"
thread":"1:0x7b17921306c0","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS
","category_id":7,"verbose_level":"DEBUG_1","verbose_level_id":1,"msg":"saving checkpoint snapshot nIn: 5
2, snapshot max: 52 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base
write gen: 1'}}}}
mongod | [{"t":{"$date":"2026-02-13T14:08:58.303+00:00"},"s":"I", "c":"WTCHKPT", "id":22430,
"ctx":{"CheckpointInt","msg":{"WireTiger message","attr":{"message":{"ts_sec":1770991738,"ts_usec":303804,"
thread":"1:0x7b17921306c0","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS
","category_id":7,"verbose_level":"DEBUG_1","verbose_level_id":1,"msg":"saving checkpoint snapshot nIn: 5
6, snapshot max: 56 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base
write gen: 1'}}}}
grpc-service | [gRPC] Creating item: id=2, name=Test 2
grpc-service | [gRPC] Item created successfully: 2
rest-service | 172.18.0.1 - - [13/Feb/2026 14:09:24] "POST /items HTTP/1.1" 201 -
rest-service | 172.18.0.1 - - [13/Feb/2026 14:09:27] "GET /health HTTP/1.1" 200 -
```

CAP Reflection

- Consistency vs Availability: New writes are blocked entirely.
- Partition Tolerance: If the network to the gRPC container is lost, the REST service tries to remain available but eventually triggers a fallback or circuit break.
- When connection is lost, the service fails fast and then resets to ‘half-open’ after timeout.